



# Facial Expression Recognition with Vision Transformers

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# Motivation



## Facial expressions

universal non-verbal communication



## Applications

healthcare, HCI, security, education



## Challenges

intra-class variation, occlusions, low-res images, imbalanced classes

# Previous Approaches

**Handcrafted  
features**

LBP, HOG +  
SVM

**CNNs**

strong but  
limited in  
long-range  
dependencies

**Vision  
Transformers**

capture global  
context with  
self-attention

# Project Goal

Apply transfer learning

Adapt model architecture

Evaluate performance

# Dataset

Dataset: FER2013

- 35,887 grayscale 48×48 images
- 7 classes: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral
- The training set consists of 28,709 examples, the validation set consists of 3,589 examples and the public test set consists of 3,589 examples.
- The data is imbalanced.



# Methods



Backbone | ViTMAE pretrained on ImageNet



Loss | Cross-Entropy



Optimizer | AdamW (lr=3e-5)



Scheduler | Cosine Annealing



Augmentation | resize, crop, flip, normalize

# Architecture

Patch embedding ( $16 \times 16$ )

Positional encoding

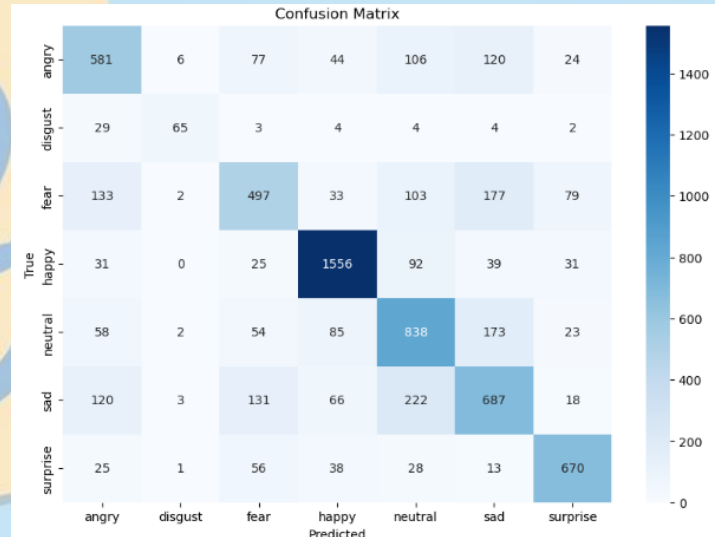
12 Transformer encoder blocks

[CLS] token  $\rightarrow$  classifier head (7 outputs)



# Results

```
[FT 78/80] loss=0.7907 val_acc=64.29  
[FT 79/80] loss=0.8128 val_acc=65.09  
[FT 80/80] loss=0.8173 val_acc=63.80
```



|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| angry        | 0.5947    | 0.6065 | 0.6005   | 958     |
| disgust      | 0.8228    | 0.5856 | 0.6842   | 111     |
| fear         | 0.5896    | 0.4854 | 0.5324   | 1024    |
| happy        | 0.8521    | 0.8771 | 0.8644   | 1774    |
| neutral      | 0.6016    | 0.6796 | 0.6382   | 1233    |
| sad          | 0.5664    | 0.5509 | 0.5585   | 1247    |
| surprise     | 0.7910    | 0.8063 | 0.7986   | 831     |
| accuracy     |           |        | 0.6818   | 7178    |
| macro avg    | 0.6883    | 0.6559 | 0.6681   | 7178    |
| weighted avg | 0.6801    | 0.6818 | 0.6794   | 7178    |

## Validation accuracy

Epoch 1 = 15%

Epoch 3 = 25%

Final = 65%

## Confusion matrix

good on 'Happy'/'Disgust'  
errors in 'Sad vs Neutral'

## Classification report

Underrepresented  
classes → lower recall



# Conclusion & Future Work

ViT improves FER performance vs baselines  
Transfer learning effective with small datasets

Future: larger datasets, hybrid models, multimodal approaches

The background is a solid light blue color. Scattered across it are several yellow sticky notes of various sizes and orientations. Each sticky note has a simple blue smiley face drawn on it. The smiley faces include eyes (represented by two dots or two curved lines) and a mouth (represented by a curved line). The notes are slightly overlapping and have a subtle drop shadow, giving them a three-dimensional appearance.

Thanks for watching!